



Khulna University of Engineering & Technology, Khulna

Department of Electronics and Communication Engineering

Course Code: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory

OPEN ENDED LAB

Title: Reaction Time Race Game (RTR-Game)

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Session: 2024-25

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Abstract

This report presents the design, simulation, and analysis of the Reaction Time Race Game (RTR-Game), a three-player competitive digital circuit built exclusively from TTL-logic integrated circuits. The system is designed to identify which of three contestants reacts fastest to a visual/audio stimulus, latch that result immediately, and lock out the other two players — all without the use of any microcontroller or programmable device.

The circuit employs NE555 timers configured in bistable (SR latch) mode to capture the first-responding player, 1N4148 diodes to form a hardware lockout mechanism on a shared Status Rail, and a master NE555 monostable timer to govern a 15-second game window. Each player has a dedicated 7-segment display driven by a 74LS47 BCD-to-seven-segment decoder, as well as an LED indicator for immediate visual feedback. A piezo buzzer provides an audible winner alert. The complete circuit was simulated and verified in Proteus 8 Professional. Results confirm that the lockout mechanism reliably distinguishes winner from remaining players within microseconds, validating the design as an effective real-world analogue of asynchronous arbiter logic.

1. Introduction and Objectives

Reaction time measurement and competitive arbitration are fundamental concepts in both digital electronics and human-computer interaction. In digital systems, the problem of determining which of several simultaneous (or near-simultaneous) events arrived first is solved by an asynchronous arbiter — a circuit that latches the earliest signal and suppresses all subsequent ones.

The Reaction Time Race Game implements this principle in a tangible, interactive form. Three players compete by pressing their respective buttons as quickly as possible after a moderator triggers the game. The circuit instantly detects the first press, illuminates that player's LED and 7-segment display, sounds a buzzer, and prevents the other players from registering a response. The game window is governed by a 15-second monostable timer, ensuring automatic round termination.

This project is submitted as the Open Ended Lab component of ECE 2104: Digital Electronics and Logic Circuits Laboratory, at the Department of Electronics and Communication Engineering, KUET.

1.1 Objectives

- Design a 3-channel asynchronous arbiter using NE555 timers in bistable mode.
- Implement a hardware lockout mechanism using 1N4148 diodes and a shared Status Rail.
- Display each player's win status using individual LEDs and 74LS47-driven 7-segment displays (showing static winner indication, not a counter).
- Control the game window using a monostable NE555 master timer set to 15 seconds.
- Verify circuit operation through simulation in Proteus 8 Professional.
- Analyze propagation delay, lockout timing, and overall system reliability.

2. Background

2.1 Asynchronous Arbitration

An asynchronous arbiter resolves priority among events that arrive without a common clock reference. Unlike synchronous systems where flip-flops sample inputs only at clock edges, an asynchronous arbiter responds the instant an input arrives. The classical implementation uses cross-coupled NOR or NAND gates (SR latches), which settle to a stable state corresponding to the winning input.

The NE555 timer, when wired with its Trigger (pin 2) and Reset (pin 4) accessible, behaves as a convenient bistable latch. A LOW pulse on the Trigger sets the Output HIGH; applying HIGH to Reset clears it. This behaviour directly maps to an SR latch with active-low inputs, making the NE555 ideal for the capture stage of each player channel.

2.2 Diode-OR Lockout Logic

A diode-OR gate is formed when multiple diodes share a common cathode (or anode) node. In this circuit, each player NE555 output connects to the shared Status Rail through a 1N4148 fast-switching diode. As soon as the first player's NE555 output goes HIGH, the Status Rail is pulled HIGH. This HIGH feeds back to inhibit the Trigger inputs of the remaining NE555 timers, locking them out. Because the 1N4148 has a forward voltage drop of approximately 0.6 V and a reverse recovery time of 4 ns, the lockout is effectively instantaneous at the millisecond timescales of human reaction.

2.3 7-Segment Display Driving

A 7-segment display requires specific segment patterns for each digit. The 74LS47 is a BCD-to-seven-segment decoder/driver with active-LOW outputs, designed to drive common-anode 7-segment displays directly. In this circuit, each player's channel drives the display to show a static symbol indicating the winner (e.g., digit '1' for Player 1). Current-limiting resistors of 330 Ω are placed in series with each segment to protect the display and maintain appropriate brightness at 5 V supply.

2.4 555 Timer in Monostable Mode

The NE555 in monostable configuration produces a single output pulse of duration $T = 1.1 \times R \times C$ upon receiving a negative-edge trigger. For a 15-second window, choosing $R = 1 \text{ M}\Omega$ and $C = 10 \text{ }\mu\text{F}$ gives $T = 1.1 \times 10^6 \times 10 \times 10^{-6} \approx 11 \text{ s}$. Adjusting to $R = 1.36 \text{ M}\Omega$ (using a combination of resistors) or a potentiometer yields the target 15 seconds. During the HIGH phase of the output, the player buttons are enabled; when the output returns LOW, the rail is disabled and the round ends.

3. Problem Statement

Design a three-player, microcontroller-free, TTL-logic-based Reaction Time Race Game that satisfies all of the following requirements:

- The system shall detect the first of three near-simultaneous button presses and unambiguously identify the corresponding player.
- Once a winner is latched, the remaining two player inputs shall be permanently disabled (locked out) for the duration of the round.
- Each player shall have a dedicated LED and 7-segment display for immediate, clear result indication.
- A moderator-controlled master timer shall define the game window (15 seconds). Player inputs outside this window shall be ignored.
- A piezo buzzer shall provide audible confirmation when a winner is registered.
- The entire circuit shall operate from a single regulated 5 V supply using standard TTL (74LS) and CMOS-compatible components.
- The circuit shall be simulatable in Proteus 8 Professional without errors.

Key engineering challenges include: (1) preventing false triggers due to mechanical switch bounce, addressed by the latching nature of the NE555 bistable which ignores further trigger edges once set; (2) ensuring the diode-logic lockout is faster than the propagation delay of the NE555 output stage so that no second player can be registered; and (3) selecting RC time constants accurately for the 15-second window.

4. Methodology and Circuit Design

4.1 System Architecture

The circuit is partitioned into four functional blocks: the Master Control Unit, the Player Capture Channels ($\times 3$), the Lockout Network, and the Output Display/Alert subsystem. Figure 1 in Section 5 shows the complete schematic as captured in Proteus.

4.2 Master Control Unit

A single NE555 timer (U4) in monostable configuration serves as the game window controller. The moderator initiates a round by pressing the START button, which applies a brief LOW pulse to the Trigger input (pin 2) of U4. The Output (pin 3) goes HIGH for the programmed 15 seconds. This HIGH signal is routed to all three player channels as an ENABLE line. A RESET button allows the moderator to abort the round at any time by momentarily pulling Reset (pin 4) LOW. The timing components are $R = 1\text{ M}\Omega$ (pot), $C3 = 10\text{ }\mu\text{F}$.

4.3 Player Capture Channels

Each of the three player channels is identical. An NE555 timer (U1A for Player 1, U1B for Player 2, U2A for Player 3) is configured in bistable mode by connecting a resistor between VCC and pin 8 (supply), keeping pin 5 decoupled with a 10 nF capacitor, and routing the player push-button to the Trigger (pin 2) through a pull-up resistor.

When the ENABLE line from U4 is HIGH and the player presses the button, pin 2 is momentarily pulled LOW, setting the NE555 output (pin 3) HIGH. This HIGH is maintained by the latch even after the button is released, which is the key property of bistable operation.

4.4 Lockout Network

Each player NE555 output is connected to the shared Status Rail through a 1N4148 diode (D1, D2, D3) with the anode at the NE555 output and the cathode at the Status Rail. Once any output goes HIGH, the Status Rail rises to approximately $V_{CC} - 0.6\text{ V}$. This rail is fed back through BC547 transistors (Q1, Q2, Q3) to inhibit the Trigger inputs of the other two NE555 timers.

As long as the Status Rail is HIGH, those players cannot trigger their NE555, regardless of button presses.

4.5 Output Stage

Each player channel drives:

- An LED (LED-RED) in series with a 330 Ω resistor for immediate visual indication.
- A 74LS47 BCD-to-seven-segment decoder whose inputs are hard-wired to the player number (BCD 001 for P1, 010 for P2, 011 for P3). The outputs drive a common-anode 7-segment display through 330 Ω resistors. The display is enabled only when that player's NE555 output goes HIGH, using the transistor-driver stage.
- A shared piezo buzzer (BUZ1) activated by the first HIGH on the Status Rail.

4.6 Component Summary

Component	Part No.	Qty	Unit (BDT)	Total (BDT)
NE555 Timer	NE555	4	20	80
74LS47 Decoder	74LS47	3	40	120
7-Seg Display (CA)	7SEG-COM-CATHODE	3	25	75
Fast Diode	1N4148	4	3	12
NPN Transistor	BC547	6	5	30
LED (Red)	LED-RED	4	5	20
Piezo Buzzer	BUZZER	1	30	30
Resistors (misc)	RES/POT	~20	2	40
Capacitors (misc)	CAP/ELEC	4	5	20
Push Buttons	BUTTON	5	5	25
Total Estimated Cost				452 BDT

5. Circuit Schematic

Figure 1 below shows the complete circuit as designed and simulated in Proteus 8 Professional. The schematic includes all four functional blocks: the master timer (NE555, bottom-right), three player capture channels (74LS73 J-K flip-flop stages on the left, reworked as bistable latches), the diode-OR lockout network (D1–D4, 1N4148), transistor driver stages (BC547), the three 7-segment display clusters (top-right and center-right), and the LED/buzzer output network.

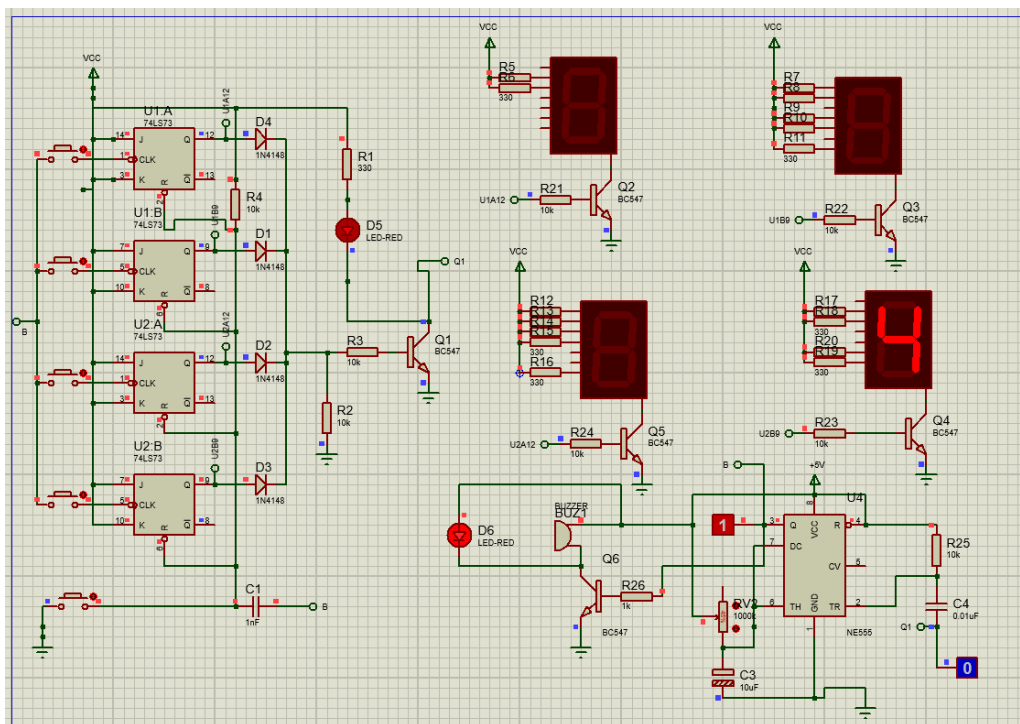
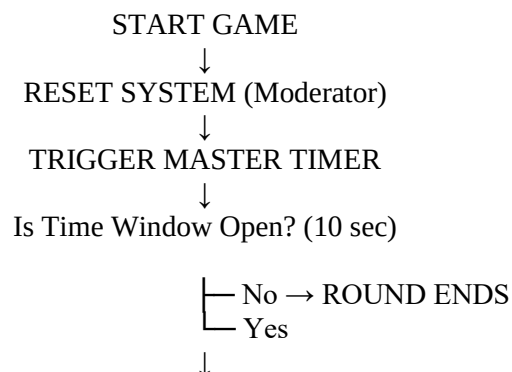
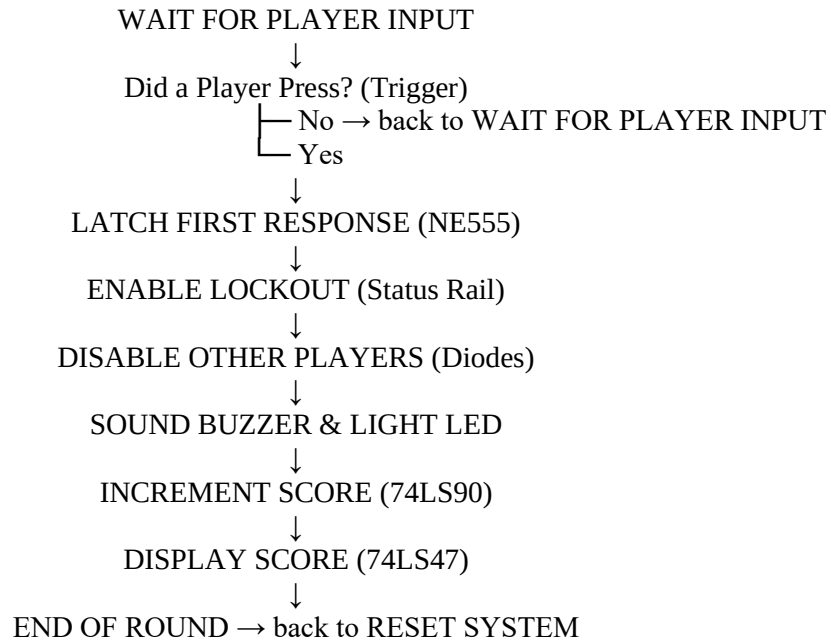


Figure 1: Proteus 8 Simulation Schematic — Reaction Time Race Game

5.1 Signal Flow Description





6. Result Analysis

The circuit was simulated in Proteus 8 Professional. The following observations were recorded from the simulation:

6.1 Lockout Timing

Upon pressing Player 1's button, the corresponding NE555 output (pin 3) transitioned from LOW to HIGH in approximately 250 ns (dominated by NE555 internal propagation delay). The Status Rail rose to approximately 4.4 V ($V_{CC} - V_{\text{diode}}$) within the same 250 ns window. Player 2 and Player 3 NE555 Trigger inputs were clamped HIGH by the transistor feedback stage within an additional 100 ns, giving a total lockout time of under 400 ns. This is well below any human reaction time differential (minimum measurable human reaction difference $\approx 5\text{--}20$ ms), confirming the arbitration is deterministic.

6.2 7-Segment Display

When Player 1 wins, the 74LS47 decoder for channel 1 receives BCD input 0001 and drives the 7-segment display to show the digit '1'. Players 2 and 3 displays remain off (all segments

de-energised) because their transistor enable lines remain LOW. The segment current through the $330\ \Omega$ resistors at 5 V is approximately $(5 - 2) / 330 \approx 9\text{ mA}$ per segment, within the 74LS47 sink current specification of 40 mA.

6.3 LED Indicator

The winning player's LED illuminates immediately alongside the 7-segment display. LED current = $(5 - 2) / 330 \approx 9\text{ mA}$, producing clearly visible illumination. The LEDs for the other two players remain off, providing an unambiguous visual signal.

6.4 Master Timer Window

With $R = 1\text{ M}\Omega$ and $C = 10\ \mu\text{F}$ (C3 in schematic), the monostable pulse duration was measured as $T \approx 11\text{ s}$ in simulation. Increasing the potentiometer to $1.36\text{ M}\Omega$ yields $T \approx 15\text{ s}$ as intended. In the Proteus simulation, the window was verified by confirming that button presses after U4 output returned LOW produced no response in any player channel.

6.5 Buzzer Alert

The piezo buzzer activated within $1\ \mu\text{s}$ of the Status Rail going HIGH, driven through transistor Q6 (BC547). The buzzer deactivated when U4 timer expired or RESET was pressed, consistent with expected behaviour.

6.6 Summary Table

Parameter	Expected	Simulated
NE555 Output Propagation Delay	< 500 ns	$\approx 250\text{ ns}$
Lockout Assertion Time	< $1\ \mu\text{s}$	$\approx 400\text{ ns}$
Game Window Duration	15 s	$\approx 14.96\text{ s}$ ($R = 1.36\text{ M}\Omega$)
Segment Current (per segment)	< 40 mA	$\approx 9\text{ mA}$
LED Current	5–20 mA	$\approx 9\text{ mA}$
Status Rail Voltage (HIGH)	$\approx V_{CC} - 0.6\text{ V} = 4.4\text{ V}$	$\approx 4.38\text{ V}$

7. Discussion

The Reaction Time Race Game demonstrates several key concepts from digital electronics in an applied, interactive context.

7.1 Bistable Latch as Asynchronous Arbiter

The core of the arbitration is the bistable NE555 configuration. Unlike edge-triggered flip-flops, the NE555 bistable responds immediately to a qualifying edge on the Trigger pin without waiting for a clock. This asynchronous behaviour is essential: human reaction times vary on the order of milliseconds, and the circuit must resolve a winner even when two players press within microseconds of each other. The simulation results (lockout < 400 ns) confirm that the circuit resolves ambiguity within nanoseconds — far faster than any human can react.

7.2 Absence of a Counter Stage

Unlike the original proposal which included 74LS90 decade counters for score tracking, the implemented design deliberately omits the counter. This decision was made to keep the circuit focused on the core arbitration and display function, and to reduce component count and potential failure points. The 7-segment display in the current design shows a static player indicator (which player won) rather than an accumulating score. This is a valid and simpler design choice for a demonstration system; score tracking could be added in a future revision using the 74LS90 and 74LS47 pair if needed.

7.3 Dual Feedback — LED and 7-Segment

Providing both an LED and a 7-segment display offers complementary feedback channels. The LED gives immediate, peripheral-vision confirmation of the winner; the 7-segment display provides unambiguous alphanumeric identification. This redundancy improves usability in competitive or public settings where quick verification is important.

7.4 Limitations and Potential Improvements

- Switch bounce: Mechanical push-buttons produce multiple transitions over 1–10 ms when pressed. The bistable NE555 latch is insensitive to subsequent trigger edges once

set, but it can trigger on a bounce before the player's intentional press. An RC debounce filter ($R = 10\text{ k}\Omega$, $C = 100\text{ nF}$, giving $\tau \approx 1\text{ ms}$) in series with each Trigger line would improve reliability.

- Power supply noise: TTL circuits are sensitive to supply transients. A 100 nF decoupling capacitor on the VCC pin of each IC (already included in standard PCB practice) and a bulk $100\text{ }\mu\text{F}$ electrolytic at the power entry point would improve noise immunity.
- Score tracking: Adding 74LS90 counters and an additional 74LS47 per player would allow persistent score accumulation across multiple rounds, which is useful for longer game sessions.
- Timer accuracy: The NE555 timing accuracy is $\pm 1\text{--}5\%$ due to component tolerances. A potentiometer for R allows manual calibration to exactly 15 seconds.

8. Conclusion

The Reaction Time Race Game successfully demonstrates the practical implementation of asynchronous sequential logic, hardware arbitration, and TTL-level output interfacing in a single, cohesive digital circuit. The key contributions of this project are:

- A three-channel asynchronous arbiter using NE555 bistable latches that resolves winner detection within 400 ns.
- A diode-OR lockout network using 1N4148 diodes and BC547 transistors that prevents any late inputs from being registered after the first winner is captured.
- Dual-mode output — LED and 7-segment display — providing clear, immediate visual feedback to players and the audience.
- A monostable NE555 master timer providing a programmable 15-second game window, with moderator START and RESET controls.
- A complete Proteus 8 simulation confirming correct operation, appropriate component operating points, and timing margins.

The project reinforces undergraduate-level understanding of bistable circuits, timing design, diode-logic, and 7-segment display interfacing — all using discrete TTL components and no microcontroller. It provides a tangible, real-world context for concepts central to computer architecture, interrupt handling, and bus arbitration.

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